

Question				Marking details	Marks Available																	
					AO1	AO2	AO3	Total	Maths	Prac												
2	(a)			genetic material in the epithelial cell is DNA (1) Any 3 (x1) from: <table border="1"><tr><td>DNA/ epithelial cell</td><td>RNA/ virus</td></tr><tr><td>double stranded/double helix</td><td>single stranded (1)</td></tr><tr><td>deoxyribose</td><td>ribose (1)</td></tr><tr><td>contains thymine</td><td>contains uracil (1)</td></tr><tr><td>Larger/ longer molecule</td><td>Smaller/ shorter molecule (1)</td></tr><tr><td>(Genetic material contained in nucleus)</td><td>Genetic material not {contained in nucleus/ membrane bound} (1)</td></tr></table> Answers must be comparative (apart from fifth MP) Reference to prokaryotic cells is neutral	DNA/ epithelial cell	RNA/ virus	double stranded/double helix	single stranded (1)	deoxyribose	ribose (1)	contains thymine	contains uracil (1)	Larger/ longer molecule	Smaller/ shorter molecule (1)	(Genetic material contained in nucleus)	Genetic material not {contained in nucleus/ membrane bound} (1)	4			4		
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	(b)	(i)		{Endocytosis/phagocytosis} Accept micropinocytosis	1			1														
		(ii)		Any four (x1) from: 1. (Viral) RNA acts as a template (1) 2. (Free RNA nucleotides attach to their) complementary base pairs (1) Reject DNA 3. Adenine with uracil <u>and</u> cytosine with guanine (1) Accept A:U, C:G 4. RNA polymerase catalyses the formation of the new RNA molecule (1) 5. Forming the {sugar phosphate backbone/phosphodiester bond} (1)		4		4														

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		(iii)		Any five (x1) from: - Translation (1) - mRNA attaches to a ribosome (1) - tRNA brings a (specific) amino acid (1) - Complementary base pairing between mRNA codon and tRNA anticodon/ codon anticodon interaction (or description of) (1) - Ribosome has room for two tRNA molecules/Brings two amino acids into close proximity/ or description of (1) - Condensation reaction occurs/peptide bond formed (between amino acids)(1) - Ribosome moves to the next codon (or description of)/ reference to stop codon (1) - Post translation processing/ or description of e.g. folding of polypeptide chain role of Golgi(1)	5			5		
		(iv)		It does not possess {any ribosomes/tRNA / the organelles for protein synthesis} (1) Reject: organelles unqualified All other organelles = neutral It does not possess mitochondria / cannot produce its own ATP for the process (1)		2		2		
				Question 2 total	10	6	0	16	0	0